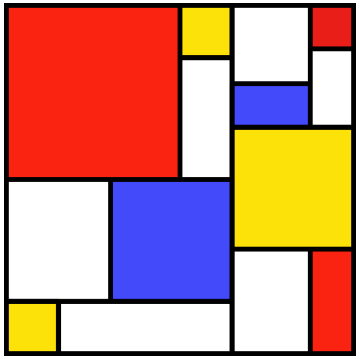
SLiM

Workshop Series



non-Wright–Fisher models

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WF versus nonWF

- SLiM supports two different model types:
  - **WF**: Wright–Fisher
  - **nonWF**: non-Wright–Fisher
- Differences:
  - tick cycles
  - offspring generation
  - population regulation
  - age structure
  - fitness models
  - demography
  - mate choice
  - migration

WF

0. Execution of `first()` events
1. Execution of `early()` events
2. Generation of offspring:
   - 2.1. Choose source subpop
   - 2.2. Choose parent 1
   - 2.3. Choose parent 2 (`mateChoice()` callbacks)
   - 2.4. Generate the offspring (including `mutation()` and `recombination()` callbacks)
   - 2.5. Suppress/modify child (`modifyChild()` callbacks)
3. Removal of fixed mutations
4. Offspring become parents
5. Execution of `late()` events
6. Fitness recalculation using `mutationEffect()` and `fitnessEffect()` callbacks
7. Tick/cycle count increment

WF Models

- Population size is a parameter
- Population regulation is automatic
- Fitness affects mating probability
- Selection is soft (relative fitness)
- Non-overlapping generations
- No age structure
- Migration is due to parameters

nonWF Models

- Population size is emergent
- Population regulation is scripted
- Fitness affects viability/survival
- Selection is hard (absolute fitness)
- Generations can overlap
- Age structure is emergent
- Migration is scripted

nonWF

0. Execution of `first()` events

1. Generation of offspring:
   - 1.1. Call `reproduction()` callbacks for individuals
   - 1.2. The callback(s) make calls requesting offspring
   - 1.3. Generate the offspring (including `mutation()` and `recombination()` callbacks)
   - 1.4. Suppress/modify child (`modifyChild()` callbacks)

2. Execution of `early()` events

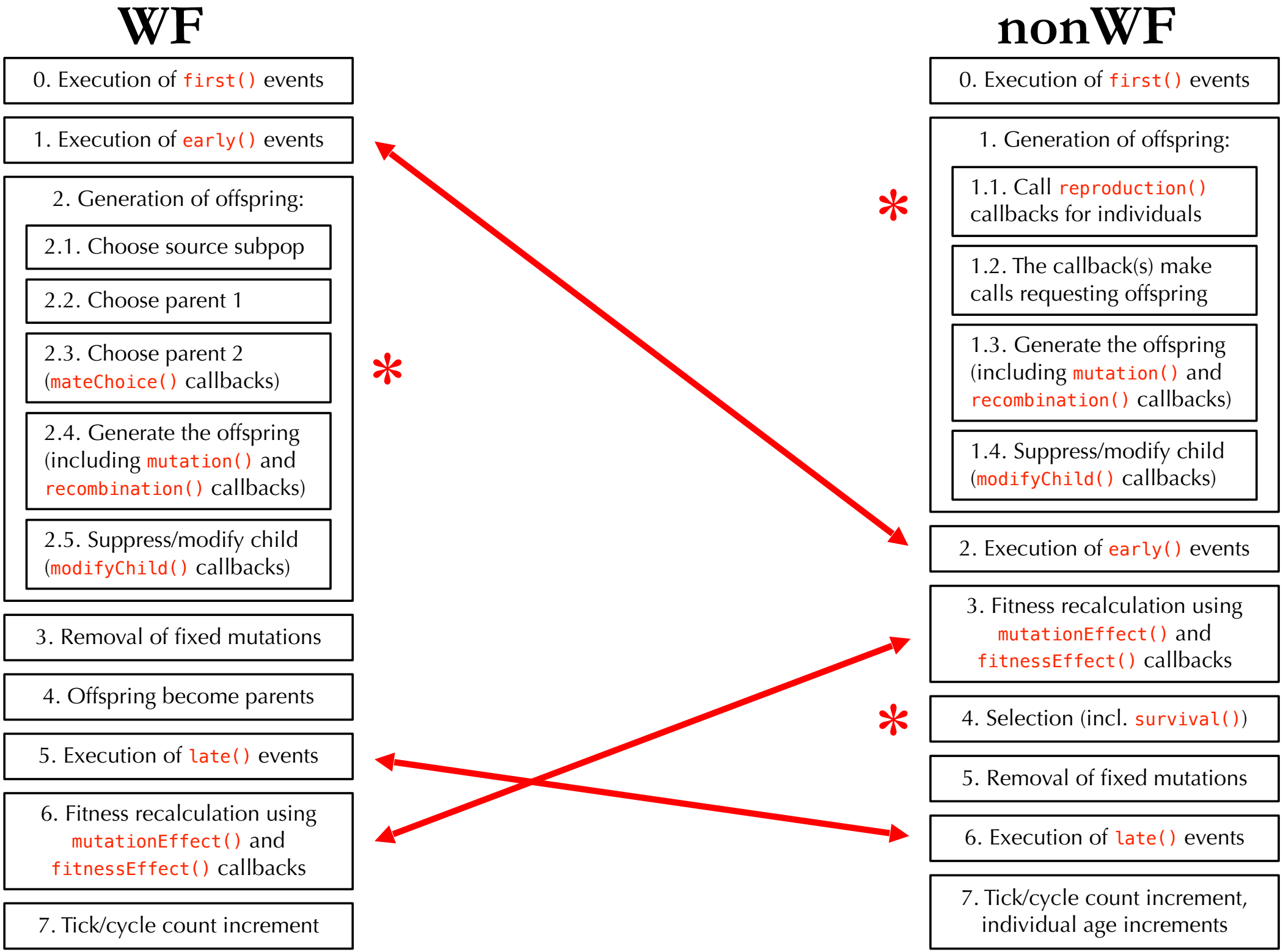
3. Fitness recalculation using `mutationEffect()` and `fitnessEffect()` callbacks

4. Selection (incl. `survival()`)

5. Removal of fixed mutations

6. Execution of `late()` events

7. Tick/cycle count increment, individual age increments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*

nonWF Models

- Need to declare the model as nonWF
  - `initializeSLiMModelType("nonWF")`
- Need to consider substitution explicitly
  - `convertToSubstitution` defaults to `F`
- Need a `reproduction()` callback
  - generates offspring for one *focal individual*
  - does so by calling `addCrossed()` etc.
- Need to impose population regulation
  - typically density-dependence in `early()`

The `initialize()` callback

```
// set up a simple neutral nonWF simulation
initialize() {
    initializeSLiMModelType("nonWF");
    defineConstant("K", 500); // carrying capacity

    // neutral mutations, which are allowed to fix
    initializeMutationType("m1", 0.5, "f", 0.0);
    m1.convertToSubstitution = T;

    initializeGenomicElementType("g1", m1, 1.0);
    initializeGenomicElement(g1, 0, 99999);
    initializeMutationRate(1e-7);
    initializeRecombinationRate(1e-8);
}
```

- Calls `initializeSLiMModelType("nonWF")`
- Defines a carrying capacity, `K`
- Sets `convertToSubstitution=T` for `m1`

The `reproduction()` callback

```
// each individual reproduces itself once
reproduction() {
    subpop.addCrossed(individual, p1.sampleIndividuals(1));
}
```

- Required in nonWF models
- Called during offspring generation
- Called once for each *focal individual*
- Does all reproduction for that individual
- Replaces the `mateChoice()` callback
- Partly replaces `modifyChild()`

The nonWF model's tick cycle

0. Execution of `first()` events

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2. Execution of `early()` events

3. Fitness recalculation using `mutationEffect()` and `fitnessEffect()` callbacks

4. Selection (incl. `survival()`)

5. Removal of fixed mutations

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`reproduction()` callbacks

```
reproduction([<subpopID>[, <sex>]])
{
    ...
}
```

- apply to *one or all* subpops
- apply to *one or all* sexes
- **pseudo-parameters:**
  - `individual`
  - `subpop`
- no return value

Population regulation

```
// provide density-dependent selection
early() {
    p1.fitnessScaling = K / p1.individualCount;
}
```

- There is no set population size
- Population size is regulated by **mortality**
- Mortality depends upon **fitness**
- Various mechanisms can operate:
  - resource availability
  - predation
  - territorial behavior
  - natural disasters

A complete nonWF model

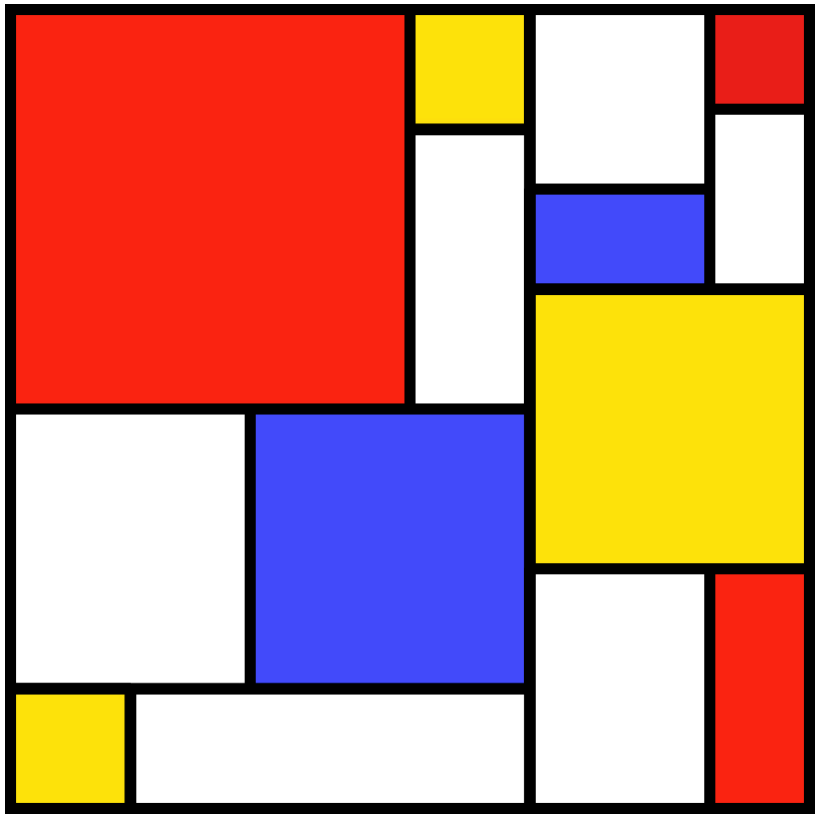
```
// set up a simple neutral nonWF simulation
initialize() {
	initializeSLiMModelType("nonWF");
	defineConstant("K", 500);  // carrying capacity

	// neutral mutations, which are allowed to fix
	initializeMutationType("m1", 0.5, "f", 0.0);
	m1.convertToSubstitution = T;

	initializeGenomicElementType("g1", m1, 1.0);
	initializeGenomicElement(g1, 0, 99999);
	initializeMutationRate(1e-7);
	initializeRecombinationRate(1e-8);
}

// each individual reproduces itself once
reproduction() {
	subpop.addCrossed(individual, p1.sampleIndividuals(1));
}

// create an initial population of 10 individuals
1 early() {
	sim.addSubpop("p1", 10);
}

// provide density-dependent selection
early() {
	p1.fitnessScaling = K / p1.individualCount;
}

2000 late() { sim.outputFixedMutations(); }
```

Fitness in nonWF models

- In WF models:
  - fitness is *relative*, selection is *soft*
  - fitness influences **mating success**

Fitness in nonWF models

- In WF models:
  - fitness is *relative*, selection is *soft*
  - fitness influences **mating success**
- In nonWF models:
  - fitness is *absolute*, selection is *hard*
  - fitness influences **mortality**
  - survival probability is capped at 100%
  - beneficial mutations behave unexpectedly!



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